ALKENE

121. Decolourization of alkaline $KMnO_4$ is used as a test for

(a) Aromatic hydrocarbons

(b) Olefinic hydrocarbons

(c) Acetylenic hydrocarbons

(d) Cycloalkanes

122. The reaction

$CH_2 = CH_2 + H_2 \xrightarrow[250-300^oC]{Ni} CH_3 - CH_3$ is called

(a) Wurtz's reaction

(b) Kolbe's reaction

(c) Sabatier and Senderen's reaction

(d) Carbylamine reaction

123. The alkene which on ozonolysis yields acetone is

(a) $CH_2 = CH_2$

(b) $CH_3 - CH = CH_2$

(c) $(CH_3)_2C = C(CH_3)_2$

(d) $CH_3 - CH = CH - CH_3$

124. $CH_3CH = CHCHO$ is oxidized to $CH_3CH = CHCOOH$ using

(a) Alkaline potassium permanganate

(b) Acidified potassium permanganate

(c) Selenium dioxide

(d) Osmium tetroxide

125. The order of increasing reactivity towards *HCl* of the following compounds will be

(1) $CH_2 = CH_2$

(2) $(CH_3)_2C = CH_2$

(3) $CH_3CH = CHCH_3$

(a) $1 < 2 < 3$ (b) $1 < 3 < 2$

(c) $3 < 2 < 1$ (d) $2 < 1 < 3$

126. The reagent which is used to distinguish between propene and propyne is

(a) Bromine

(b) Alkaline $KMnO_4$

(c) Ammoniacal $AgNO_3$

(d) Ozone

127. Which one of the following reactions would be the best for the formation of 2-bromobutane

(1) $CH_3CH = CHCH_2CH_3 \xrightarrow{HBr}$

(2) $CH_3CH_2CH = CH_2 \xrightarrow{HBr}$

(3) $CH_3CH = CHCH_3 \xrightarrow{Br_2}$

(4) $CH_3CH_2CH = CH_2 \xrightarrow[\text{Peroxide}]{HBr}$

(a) 1 (b) 2

(c) 3 (d) 4

128. If *HCl* is added over $CH_2 = C(CH_3)_2$ then what is formed

(a) $CH_2(Cl) = CH(CH_3)_2$

(b) $CH_2 - CH(Cl)(CH_3)_2$

(c) $CH_2 = C(CH_3)(CH_2Cl)$

(d) None of these

129. Position of double bond in an organic compound is determined by

(a) Ozonolysis

(b) Oxidation

(c) Reduction

(d) Hydrogenation

130. A gas decolourises Bayer's reagent but does not react with Tollen's reagent, this gas is

(a) Ethene (b) Ethyne

(c) Ethane (d) Methane

131. Formation of 2-butene from 2-bromobutane is according to

(a) Markowikoff's (b) Bayer

(c) Saytzeff (d) Wurtz

132. An alkene on ozonolysis gave acetaldehyde the alkene is

(a) Ethylene (b) Propene

(c) 1-butene (d) 2-butene

133. Indicate the organic structure for the product expected when 2-methyl propene is heated with acetyl chloride in presence of anhydrous zinc chloride

(a) $CH_3 - C(CH_3)(Cl) - CH_2 - CO - CH_3$

(b) $CH_3 - C(H)(CH_3) - CH_2 - CO - CH_3$

(c) $CH_3 - C(=O) - O - C(M)(M)(M)$

(d) $CH_3 - C(=O) - C(CH_3) = CH_2$

134. The reaction

$$CH_3 - C(CH_3)(OH) - CH_3 \xrightarrow{H_2SO_4} CH_3 - C(CH_3) = CH_2$$

is the example of

(a) Sulphonation

(b) Dehydration

(c) Alkylation

(d) Decomposition

135. Olefins can be hydrogenated by

(a) Zinc and HCl

(b) Nascent hydrogen

(c) Raney *Ni* and H_2

(d) Lithium hydride in ether

136. Electrolysis of cold concentrated aqueous solution of potassium succinate yields

(a) Ethane

(b) Ethyne

(c) Ethene

(d) Ethane-1, 2-diol

137. A hydrocarbon containing 2 carbon atoms gives Sabatier and Senderen's reaction but does not give precipitate with ammoniacal silver nitrate solution. The hydrocarbon in the question is

(a) Ethane

(b) Acetylene

(c) Ethylene

(d) None of these

138. The reaction

$CH_2 = CH - CH_3 + HBr \rightarrow$

$CH_3CHBr - CH_3$ is

(a) Nucleophilic addition

(b) Electrophilic addition

(c) Electrophilic substitution

(d) Free radical addition

139. What is the product of the reaction of 1, 3-butadiene with Br_2

(a) 1,4 –dibromobutene

(b) 1,2 -dibromobutene

(c) 3,4-dibromobutene

(d) 2,3-dibromo-2-butene

140. An alkene given two moles of *HCHO*, one mole of CO_2 and one mole of CH_3COCHO on ozonolysis. What is its structure

(a) $CH_2 = C = CH - CH_2 - CH_3$

(b) $CH_2 = CH - \overset{\overset{CH_3}{|}}{C}H - CH = CH_2$

(c) $CH_2 = C = \underset{\underset{CH_3}{|}}{C} - CH_3$

(d) $CH_2 = C = \overset{\overset{CH_3}{|}}{C} - CH = CH_2$